



EAST TEXAS A&M
— UNIVERSITY —

CSCI 352.61E

DIGITAL FORENSICS

System Forensics, Investigation, and Response

Lesson 2

Overview of Computer Crime

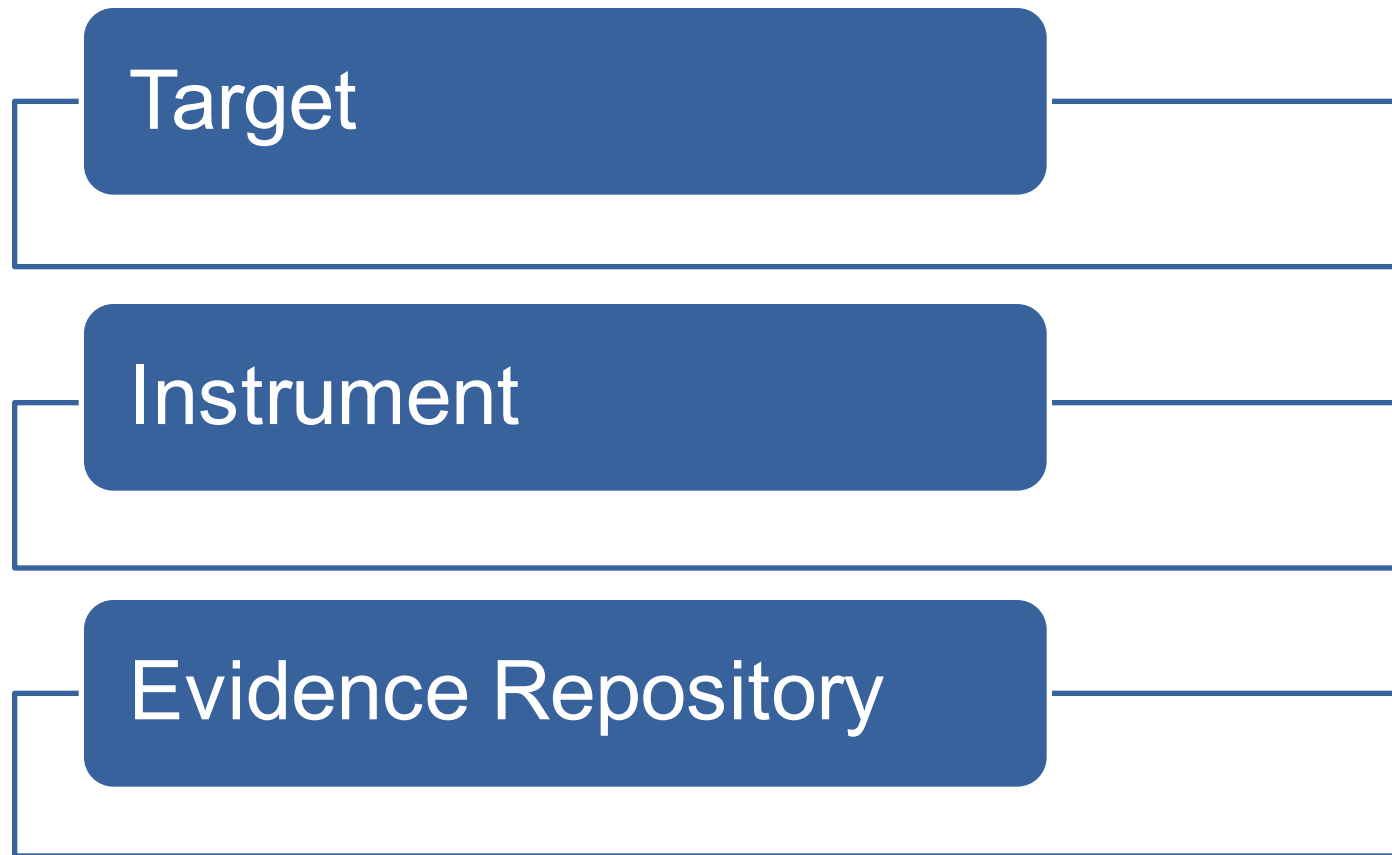
Learning Objective

- Describe various computer crimes and how they are investigated.

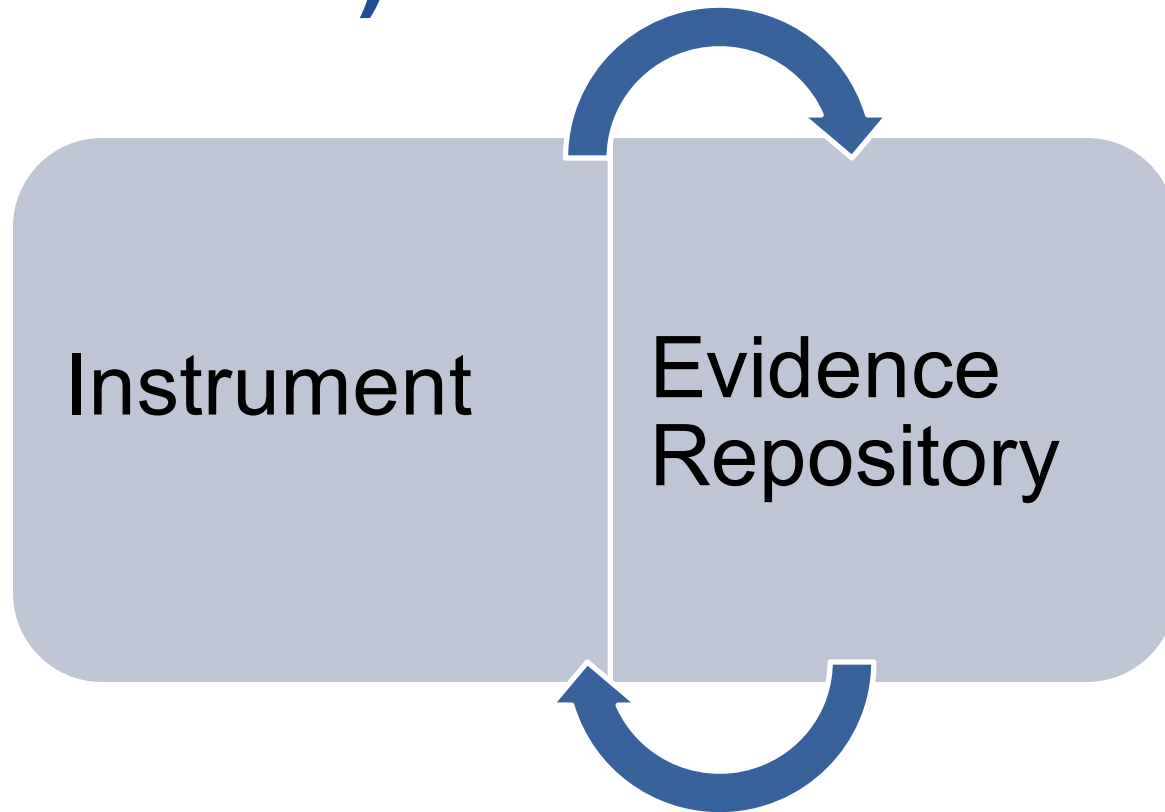
Key Concepts

- Common computer crimes
- Forensic approaches to different crimes
- Forensic strategies based on the specific crime

How Computer Crime Affects Forensics



The Role of Computers in Crimes (Continued)



Types of Crimes Involving Computers

- White-collar crimes
- Violent crimes—murder, terrorism
- Counterintelligence
- Economic espionage
- Counterfeiting
- Child pornography
- Drug dealing

Identity Theft

Use of another person's identity

Typically performed for economic gain

Methods include phishing, spyware, and discarded information

Phishing

- An attempt to trick a victim into giving up personal information
 - Usually done by emailing the victim and claiming to be from some organization a victim would trust
 - Is generally a process of reaching out to as many people as possible, hoping enough people respond
- More targeted attacks: spear phishing and whaling

Spyware

- Any software that can monitor activity on a computer
- May involve taking screenshots or perhaps logging keystrokes
- Can have legal or illegal applications

Spyware

Please respond - overdue payment

QuickBooks Invoice <auto-invoice@quickbooks.com>

Sent: Fri 2/22/2013 8:42 AM

To: chuck@chuckeasttom.com

Message Invoice_QuickBooks_22_02_2013.zip (103 KB)

Please find attached your invoices for the past months. Remit the payment by 02/25/2013 as outlines under our "Payment Terms" agreement.

Thank you for your business,

Sincerely,
Dolores Dailey

This e-mail has been sent from an automated system. PLEASE DO NOT REPLY.

The information contained in this message may be privileged, confidential and protected from disclosure. If the reader of this message is not the intended recipient, or an employee or agent responsible for delivering this message to the intended recipient, you are hereby notified that any dissemination, distribution or copying of this communication is strictly prohibited. If you have received this communication in error, please notify your representative immediately and delete this message from your computer. Thank you.

Used with permission from Microsoft

Discarded Information

- Is another method that allows a hacker to gather information about a person's identity
- Often referred to as dumpster diving
- Shred documents before throwing them out to avoid identity thief

How Identity Theft Affects Forensics

- Look for spyware on the victim's machine
- If spyware exists, search for where the spyware is sending its data
 - Periodic email with an attachment
 - A stream of packets to a server the criminal has access to
- If phishing, check email history on the victim's computer as well as the web history

Hacking

- A generic term that means breaking into a system
- Is often performed remotely

Hacking via SQL Injection

- Typical SQL statement

- `SELECT * FROM tblUsers WHERE USERNAME =
" + txtUsername.Text + ' AND PASSWORD = "
+ txtPassword.Text + "`

- Specific username and password:

- `SELECT * FROM tblUsers WHERE USERNAME =
'thisuser' AND PASSWORD = 'letmein'`

- SQL injection example:

- `SELECT * FROM tblUsers WHERE USERNAME = "
or '1' = '1' AND PASSWORD = " or '1' = '1'`

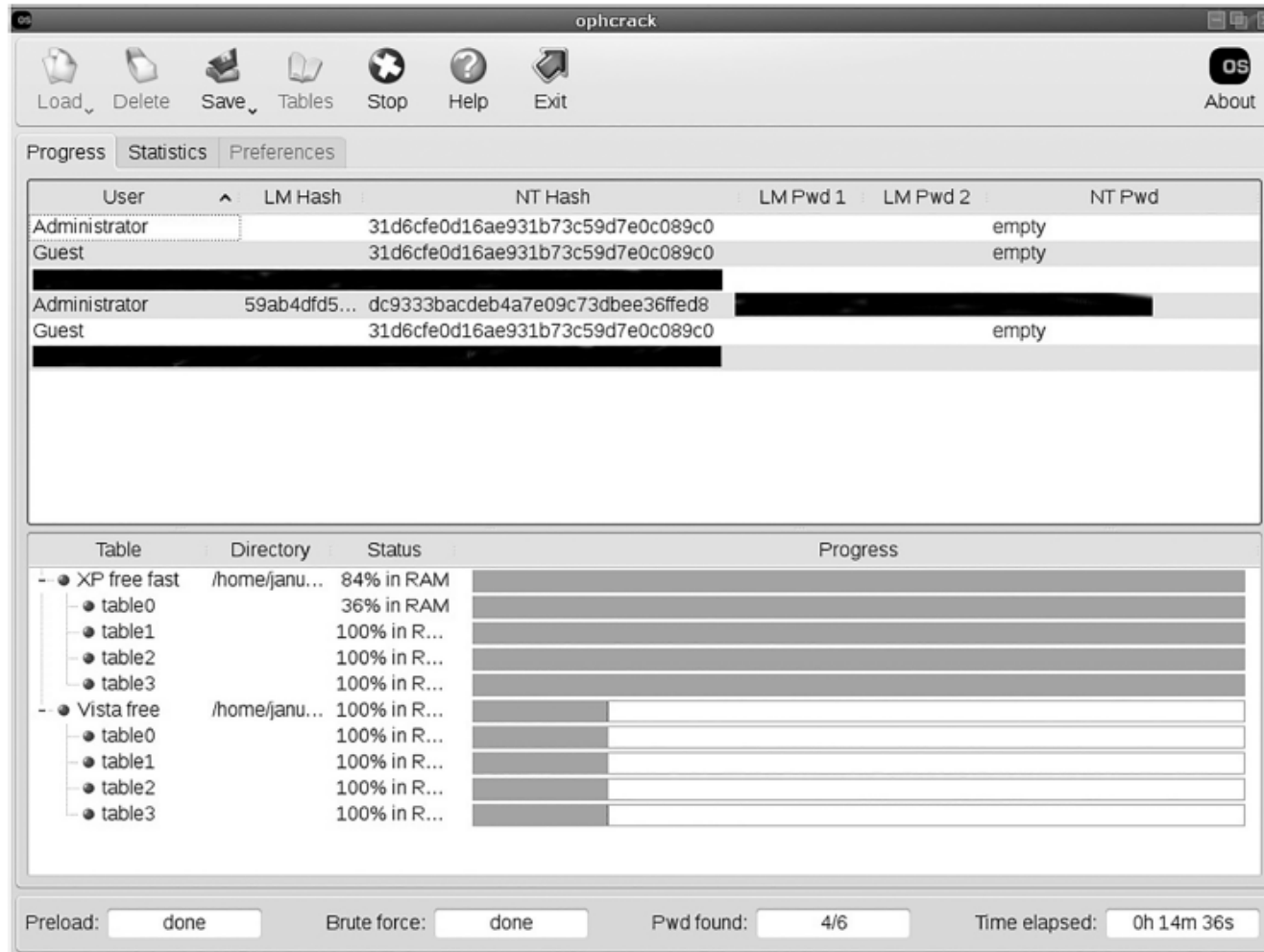
Hacking via Cross-Site Scripting

- Perpetrator seeks out someplace on target website that allows end users to post text that other users will see, such as product reviews
- Instead of posting a review or other text, the attacker posts JavaScript
- If website does not filter user input before displaying, other users navigate to this review and script executes

Hacking via Ophcrack

- When Windows boots up, system locks the SAM file so it cannot be copied or edited
- A hacker can use Ophcrack to:
 - Boot to a Linux Live CD
 - Compares rainbow table against SAM file, searching for matching passwords
- Ophcrack displays all passwords it finds

Hacking via OphCrack



Courtesy of Objectif Sécurité

Cyberstalking and Harassment

- Using electronic communications to harass or threaten another person
- Criteria for law enforcement officers:
 - Is it possible? Is the threat credible?
 - How frequent?
 - How serious?

Fraud

- Any attempt to gain financial reward through deception

Subclasses of fraud

Investment offers

Data piracy

Non-Access Computer Crimes

- Crimes that do not involve an attempt to actually access the target
- Examples:
 - Denial of service (DoS) attacks
 - Viruses
 - Logic bombs

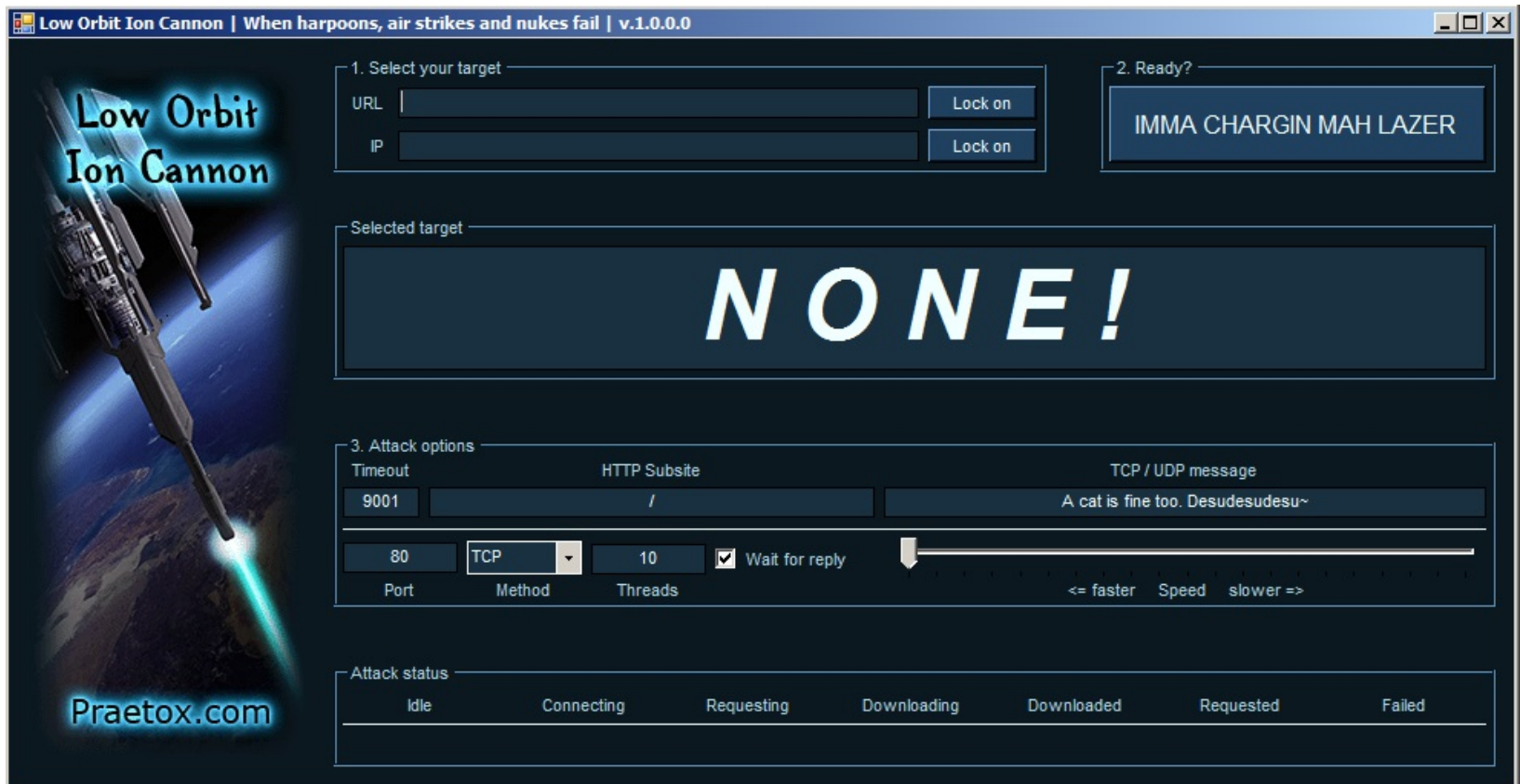
Denial of Service (DoS) Attacks

- Three-way handshake
 - Client machine sends TCP packet to server with synchronize (SYN) flag turned on
 - Server acknowledges the request to synchronize by sending back a TCP packet with two flags turned on: acknowledgment (ACK) flag and the SYN flag
 - Client responds with a single ACK flag and communications begin

Denial of Service (DoS) Attacks

- SYN flood
 - Attacker keeps sending SYN packets but never responds to the SYN/ACK it receives from the server
 - Eventually, server has so many open connections it can no longer respond to legitimate users

Low Orbit Ion Cannon



Courtesy of Praetox Technologies

Viruses

- Any software that self-replicates, like a human or animal virus
- Recent viruses:
 - FakeAV.86
 - Flame
 - Gameover ZeuS
 - Rombertik
 - Locky (ransomware)

Logic Bombs

- Malware designed to harm a system when some logical condition is reached
- Often triggered based on a specific date and time
- Possible to distribute a logic bomb via a Trojan horse

Cyberterrorism

- The unlawful use of the Internet to perform terrorist activities
- Can include large-scale disruption of computer networks
- China Eagle Union
 - Group that consists of several thousand Chinese hackers whose stated goal is to infiltrate Western computer systems

Summary

- Common computer crimes
- Forensic approaches to different crimes
- Forensic strategies based on the specific crime

Let's Play:

Identify the Crime

Question 1

- An attacker disseminates a virus from a rogue website that infects many computers.

- Answer choices:
 - Identity theft
 - Hacking
 - Cyberstalking and harassment
 - Computer fraud
 - Non-access computer crime
 - Cyberterrorism

Question 2

- An attacker sends out false emails suggesting the receiver can make a large sum of with very little investment.

- Answer choices:
 - Identity theft
 - Hacking
 - Cyberstalking and harassment
 - Computer fraud
 - Non-access computer crime
 - Cyberterrorism

Question 3

- An attacker inserts SQL commands into text boxes, often using the username and password text fields on a logon screen.

- Answer choices:
 - Identity theft
 - Hacking
 - Cyberstalking and harassment
 - Computer fraud
 - Non-access computer crime
 - Cyberterrorism

Question 4

- A criminal uses phishing to trick a victim into giving up personal information.

- Answer choices:
 - Identity theft
 - Hacking
 - Cyberstalking and harassment
 - Computer fraud
 - Non-access computer crime
 - Cyberterrorism

Question 5

- A denial of service (DoS) attack is an example of this type of this crime.

- Answer choices:
 - Identity theft
 - Hacking
 - Cyberstalking and harassment
 - Computer fraud
 - Non-access computer crime
 - Cyberterrorism

Question 6

- Data piracy is an example of this type of crime.

- Answer choices:
 - Identity theft
 - Hacking
 - Cyberstalking and harassment
 - Computer fraud
 - Non-access computer crime
 - Cyberterrorism

Question 7

- An attacker remotely accesses a power plant's computer system and inserts a logic bomb.

- Answer choices:
 - Identity theft
 - Hacking
 - Cyberstalking and harassment
 - Computer fraud
 - Non-access computer crime
 - Cyberterrorism

Question 8

- An attacker dumpster dives to look for a victim's personal information, such as in discarded mail, bills, and bank statements.

- Answer choices:
 - Identity theft
 - Hacking
 - Cyberstalking and harassment
 - Computer fraud
 - Non-access computer crime
 - Cyberterrorism

Question 9

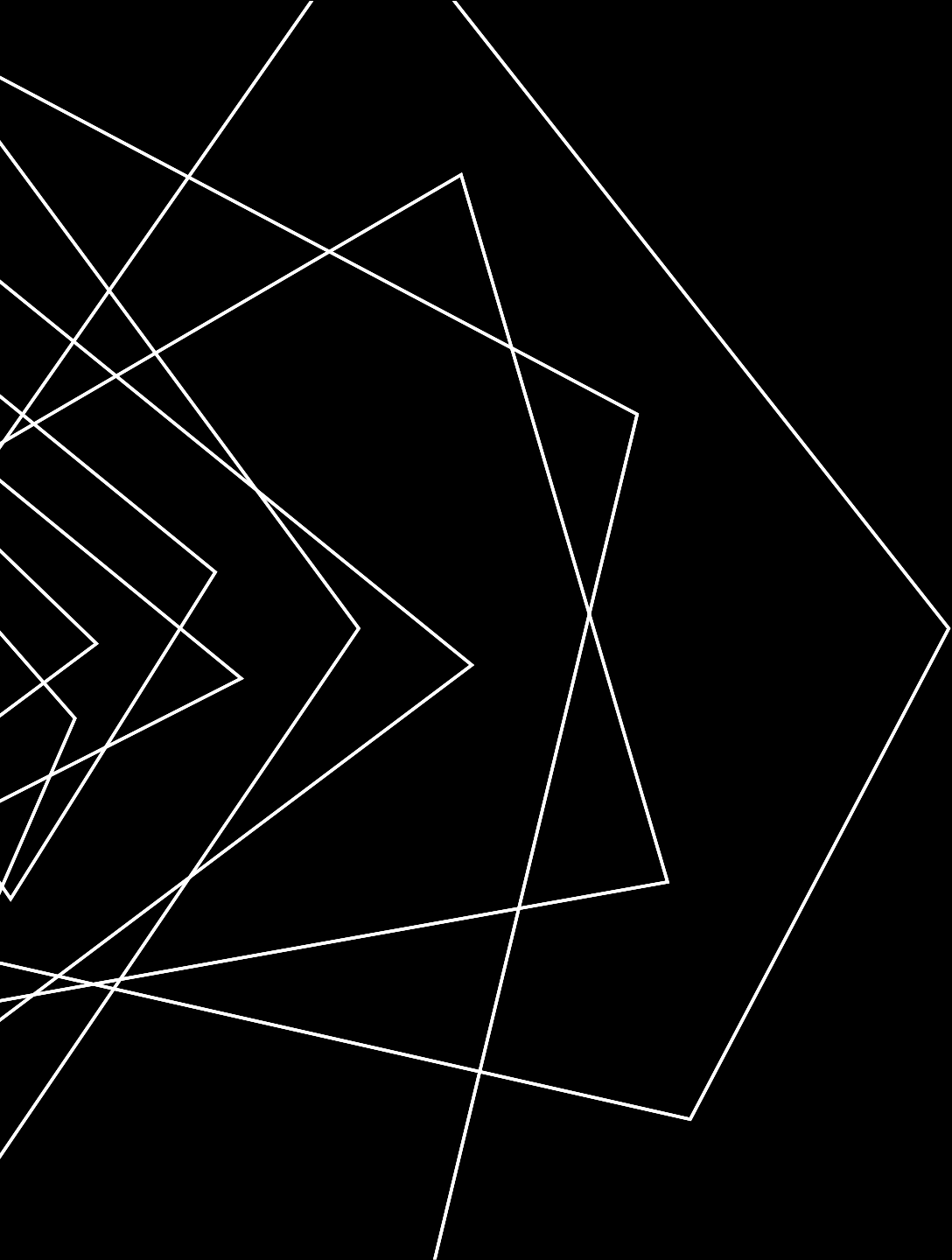
- A suspicious person in a chat room asks for your home address every time you are both online together.

- Answer choices:
 - Identity theft
 - Hacking
 - Cyberstalking and harassment
 - Computer fraud
 - Non-access computer crime
 - Cyberterrorism

Question 10

- A person from Nigeria emails you, asking to use your bank account to “park” some money temporarily.

- Answer choices:
 - Identity theft
 - Hacking
 - Cyberstalking and harassment
 - Computer fraud
 - Non-access computer crime
 - Cyberterrorism



THANK YOU

Yau Ka Cheung
Yauka.Cheung@etamu.edu
ACB1 Room 335